

University Two Mark Questions
Dynamics of Machinery

Unit – I

1. State conditions for equilibrium of two force members.
2. Define crank effort.
3. Explain the significance of inertia force analysis.
4. Single cylinder engine needs heavier size flywheel. justify.
5. State the D'Alembert's principle
6. State the function of flywheel in a prime mover.
7. What is meant by piston effort and crank effort and crank effort?
8. What is the function of flywheel in punching press?
9. Define the significance of inertia force analysis

Unit – II

1. Define damping and viscous damping.
2. What do you understand by vibration isolation?
3. Define transmissibility. (2)
4. Write short note on degrees of freedom
5. What is meant by critical damping?
6. What is meant by free and forced vibrations?
7. The piston of a steam engine moves with SHM. The crank rotates at 120rpm with a stroke of 2 m.
8. Define forced vibration and damped vibrations.
9. Obtain the natural frequency of vibration for the system when two springs are connected in series. Take $s_1 = 2000 \text{ N/m}$, $s_2 = 1500 \text{ N/m}$ and $m = 10 \text{ kg}$.
10. Define critical damping and give an application of optical damping.
11. Define the term "Logarithmic Decrement".
12. Define Damping factor..
13. What is the need for vibration isolation.

Unit – III

1. Define critical speed.
2. State natural frequency of the torsional vibration of a simple system.

3. Write the expression for natural frequency of vibration using Rayleigh's method.
4. What do you understand by torsionally equivalent shaft.
5. Define the term: whirling speed of a shaft
6. Define torsionally equivalent shaft.
7. Derive an expression for natural frequency for two and three rotor systems.
8. Differentiate longitudinal and transverse vibration.

Unit – IV

1. Define hunting and isochronous.
2. What is controlling force in governor.
3. What is the function of the governor.
4. Explain the gyroscopic effects on aircrafts.
5. In what way the centrifugal governor differ from inertia governor.
6. What is gyroscopic effect on a sea moving vessels.
7. What is meant by hunting?
8. Define gyroscopic couple.
9. How governor differs from flywheel?
10. What is gyroscopic torque?
11. How does flywheel differ from the governor.

Unit – V

1. State conditions for dynamic balancing.
2. Why complete balancing is not possible in reciprocating engine.
3. Why is partial balancing done for reciprocating.
4. Why rotating masses are to be dynamically balanced.
5. What is meant by static balancing and dynamic balancing.
6. State the effects of partial balancing in locomotives.
7. What are the types of balancing machines?
8. Why balancing is required in high speed engine?
9. Define shaking force and hammer blow.
10. Differentiate Dynamic and Static balancing.
11. Complete balancing of the reciprocating mass in a single cylinder is not possible. Justify.

DYNAMICS OF MACHINERY IMPORTANT QUESTION

2-Marks

1. State D'Alembert's principle?
2. Define piston effort
3. Define inertia torque and inertia force.
4. Define static force analysis?
5. State the function of flywheel
6. Define Turning Moment diagram
7. Draw the turning moment diagram for four stroke engine.
8. Define coefficient of fluctuation of energy.
9. Define coefficient of steadiness
10. What do you mean by fluctuation of energy?
11. Define critical damping and give an application of critical damping.
12. Differentiate longitudinal and transverse vibration.
13. What is the meaning of equivalent shaft?
14. Define damping and viscous damping.
15. What do you understand by vibration isolation?
16. Write short note on degrees of freedom.
17. What do you understand by torsionally equivalent shaft?
18. What is meant by free and forced vibrations?
19. What is amplitude reduction factor?
20. What is meant by transmissibility?

11 marks

21. The length and connecting rod of a horizontal reciprocating engine are 200mm and 1meter respectively. The crank is rotating at 400rpm. When the crank has turned 30° from the inner dead center, the difference of pressure between cover end and piston rod is 0.4 N/mm^2 . If the mass of the reciprocating parts is 100Kg and a cylinder bore is 0.4meters. Calculate (i) Inertia force (ii) Force on piston (iii) Piston effort (iv) Thrust on the side of the cylinder walls (v) Thrust in the connecting rod (vi) Crank effort.
22. During forward stroke of the piston of the double acting steam engine, the turning moment has the maximum value of 2000 N-m when the crank makes an angle of 80° with the inner dead center. During the backward stroke, the maximum turning moment is 1500 N-m when the crank makes an angle of 80° with the outer dead center. The turning moment diagram for the engine may be assumed for simplicity to be represented by two triangles. If the crank makes 100 r.p.m. and the radius of gyration of the flywheel is 1.75 m, find the coefficient of fluctuation of energy and the mass of the flywheel to keep the speed within $\pm 0.75\%$ of the mean speed. Also determine the crank angle at which the speed has its minimum and maximum values.
23. The mass of an electric motor is 120 kg and it runs at 1500 r.p.m. The armature mass is 35kg and its C.G. lies 0.5 mm from the axis of rotation. The motor is mounted on five springs of negligible damping so that the force transmitted is one-eleventh of the impressed force. Assume that the mass of the motor is equally distributed among the five springs. Determine :
 - a) Stiffness of each spring;
 - b) Dynamic force transmitted to the base at the operating speed
 - c) Natural frequency of the system

24. The mass of a single degree damped vibrating system is 7.5 kg and makes 24 free oscillations in 14 seconds when disturbed from its equilibrium position. The amplitude of vibration reduces to 0.25 of its initial value after five oscillations. Determine: (a) stiffness of the spring, (b) logarithmic decrement, and (c) damping factor. i.e. the ratio of the system damping to critical damping.
25. A single cylinder, single acting, four stroke cycle gas engine develops 20 kw at 200 rpm. The work done by the gases during the expansion stroke is 3 times the work done on the gases during compression stroke. The work done on the suction and exhaust strokes may be neglected. If the flywheel has a mass of 1000 kg and has a radius of gyration of 0.6 m, find the cycle fluctuation of energy and coefficient of fluctuation of speed.